

Where Tech Meets Heart: Your Magical Tour at XJTU

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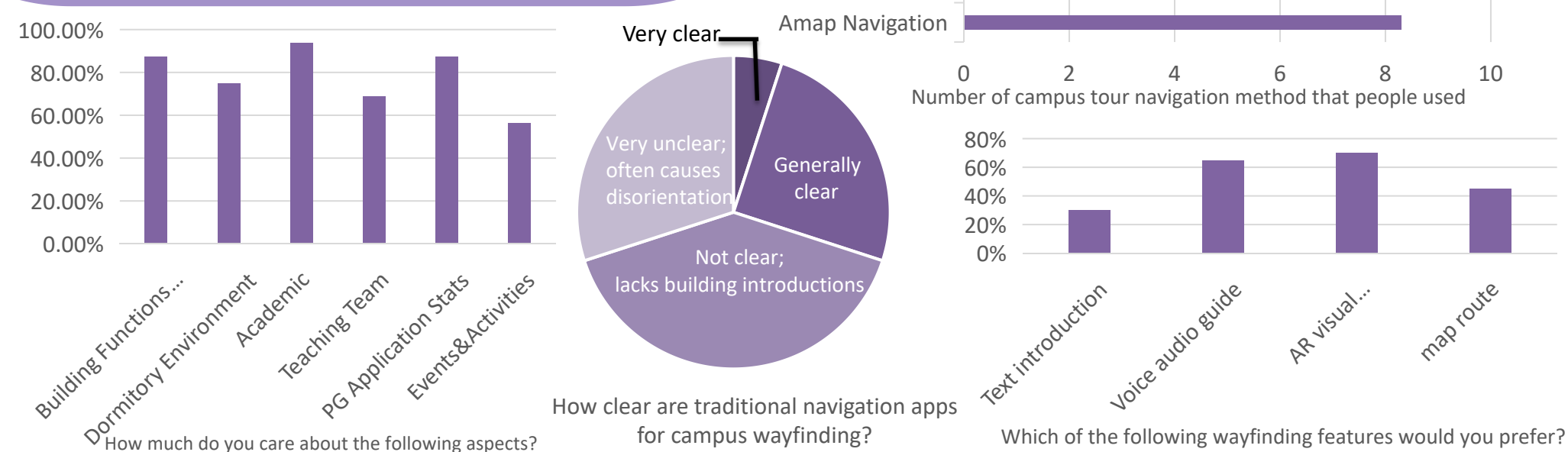


Introduction

Problems:

Wayfinding and identifying individual buildings across large university campus often imposes a substantial cognitive burden on new students, visitors, and especially prospective students and their families [1]. Unfamiliarity with campus layouts, inconsistent signage, and the sheer scale of modern university grounds can lead to confusion, disorientation, and inefficient navigation. **Without targeted and contextually appropriate guidance**, individuals frequently struggle to locate key facilities efficiently, resulting in frustration, wasted time, and a diminished overall first impression of the institution.

Requirements Discovery



We conducted a pilot test with **16 participants** (8 students + 8 parents), including task completion tests, semi-structured interviews, heuristic evaluation, and SUS usability survey.

- Core Navigation & Tracking**
Users should be able to search and route to key facilities simultaneously.
- Proximity-Based Audio Guide**
The system shall trigger automatic voice introductions when a user is within **15 meters** of a registered building.

- Academic Transparency**
Provide pop-up profiles of faculty members, curriculum highlights, and graduate outcome reports when near specific academic departments.
- "Where Tech Meets Heart"**
The system should not just be a map; it should serve as a bridge between prospective students and the authentic university lifestyle, answering the "Is it worth my Gaokao score?" question through data and social proof.

Design Solutions

Objectives:

- Find Facilities Easily**
Locate libraries, self-study rooms, dining halls, dorms, sports centers, and services.
- Learn Through Info**
Auto-play audio or text when you're nearby - campus information and tips.
- Plan Efficient Visit**
Get smart routes that save time and cover what care about most.
- Track Progress**
Collect badges for visits and discoveries, build your campus checklist.

Human-Centric:

What Makes It Different

Traditional Apps	VS	Our System
- You search for info - Static maps are hard to read - One fixed route for everyone - No context, no motivation		- Info finds you at the right place - AR + live location bring campus to life - Different style for student/parent - Badges and progress make it fun
Why People Will Love It		
Less Mental Effort Stop squinting at maps. Focus on discovering.		You're in Control Pick your interests or let AI plan the best route.
		Feels Like a Game Explore, collect, and feel proud of your journey.

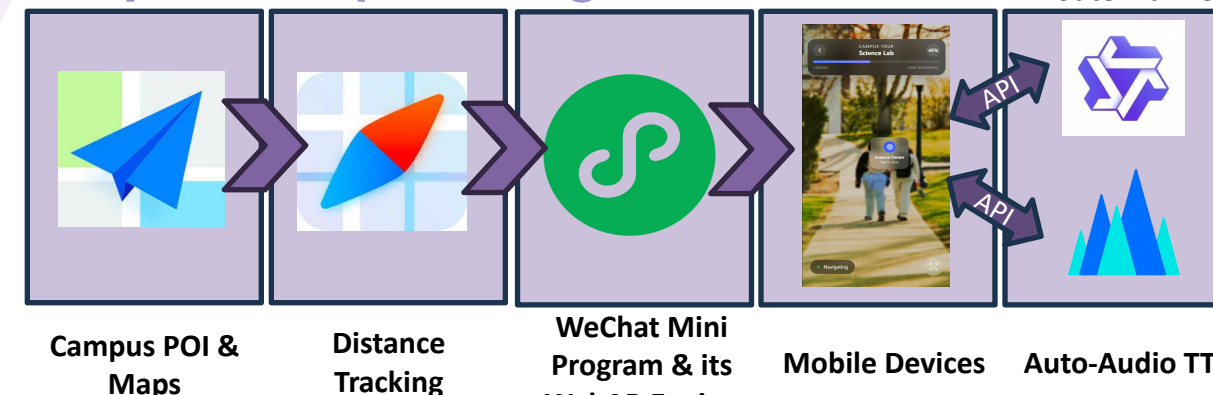
Design Concept:

- Smart Location Awareness**
 - Stories and tips appear automatically as you walk.
 - No searching, no tapping—just listen and explore.
- Dual Navigation Modes**
 - Info Mode:** Follow curated routes like "Academic Tour" or "Study Day".
 - Smart Mode:** AI recommend the best path based on Amap API.
- AR Guidance Overlay**
 - Point your camera and see arrows, building names, distance.
 - No map-reading skills needed.
- Progress & Collection**
 - Unlock badges for visiting buildings, finding completing tours.
 - Build your personal album through trip.



Figure 1 User Journey Map (1)

System operating structure:



Prototypes and Iteration

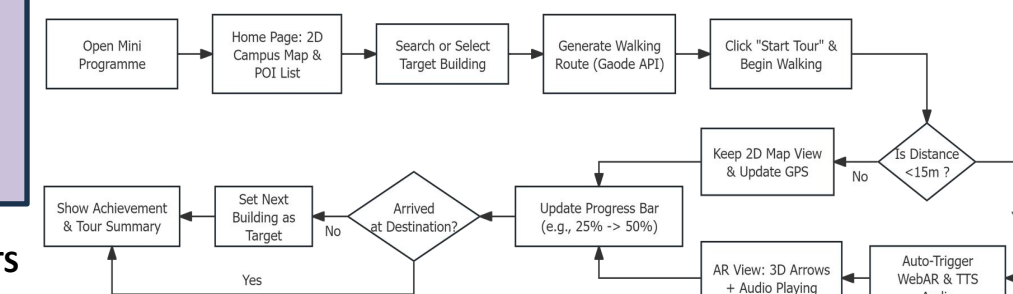


Figure 2 Early versions of flowcharts

UI design process:

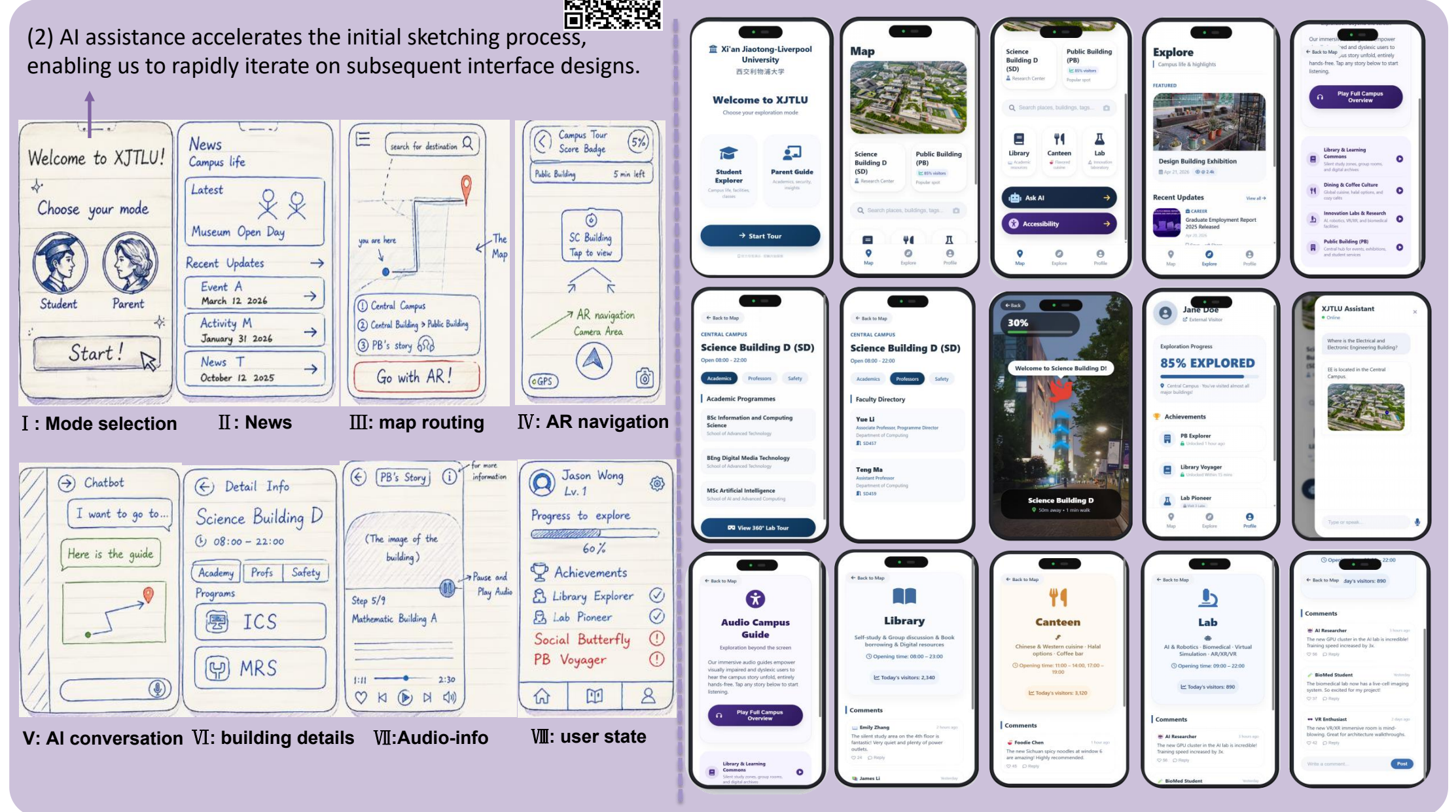


Figure 4 From hand draft to professional Figma design

Evaluation and Impact

Heuristic Evaluation:
→ Match System to Real World: AR dashboards and **15m passive voice** triggers seamlessly blend digital info with the physical XJTU campus.
→ Recognition over Recall: Campus heatmaps guide users intuitively, effectively reducing choice paralysis for first-time visitors.

- Why This Project Matters: Human-Centric Design?**
- Solve Pain Points:** Traditional tours bring cognitive overload, inflexible routes, language barriers, and a missing cultural context. Our solution shifts exploration from "finding places" to "telling stories".
 - Emotional Wellbeing & Belonging** based on **Norman's Emotional Design**: This eases new students' campus anxiety, breaks visitors' language barriers, and drives engagement via gamified progress tracking.
 - Inclusive & Accessible Design** based on **Universal Design Principles**: Hands-free audio, bilingual content, high-contrast UI for all users.

Semi-Structured Interviews:

→ Students highlighted that the "NFC real reviews" and heatmaps created a strong sense of belonging and community.
→ Parents appreciated the transparent access to academic and living facilities, reporting a significant reduction in campus-tour anxiety.
→ Both groups praised the personalized **"Dual-Mode"** for matching their distinct goals.

User Satisfaction & Feedback

Post-study questionnaires were administered to the full cohort (N=16). The system received an Overall Satisfaction Rating of 4.6/5.0, with approximately 94% of participants indicating they would recommend the tool to peers. As detailed in the feedback breakdown, users specifically highlighted the system's intuitive interface and responsiveness, validating the design's effectiveness in reducing cognitive load during campus navigation.

- Redefine Campus Experience: We prove technology serves humanity. Blending NFC/AR with storytelling, we set a new standard for university community connection.

Major Design Decision:

Design Options	Option A: Split Screen (Top half AR camera, bottom half 2D map)	Option B: AR-Dominant + Mini-map (Full-screen AR with a floating mini-map in the corner)	Option C: Contextual Dynamic Toggle (2D map default, auto-triggers full-screen AR at <15m)
Pros	Comprehensive Context: Users can simultaneously view their immediate physical surroundings and the macro route plan.	High Immersion: Maximizes visual impact and provides highly intuitive directional cues for immediate surroundings.	Right Info at the Right Time: Perfectly balances walking safety with immersive interactivity. Adheres strictly to minimalist design heuristics.
Cons	High Cognitive Load: Creates a highly cluttered mobile interface. Processing two different spatial representations while walking poses a significant safety hazard.	Loss of Global Awareness: The mini-map is too small for macro route planning. Users lose the "big picture" of the campus layout, leading to wayfinding anxiety.	High Technical Threshold: Highly dependent on the precision and low latency of the underlying GPS and haversine distance calculation engine to ensure accurate trigger timing.
Decision	Rejected	Rejected	Accepted

CONCLUSION:

This project contributes a campus navigation system that combines geofenced proactive triggers and AR visual guidance with intelligent route planning. Unlike conventional map-based tools, our design automatically **pops building information** when users approach within 15m, supported by both voice and AR feedback. Limitations include GPS accuracy under building occlusion and device performance requirements. Future work will integrate multi-source positioning such as **WiFi/Bluetooth**, and conduct user studies to evaluate spatial learning outcomes.

Conclusion & AI Acknowledgment

AI Acknowledgement:

(1) Figma, Claude Sonnet 4.5, knowledge cutoff January 2025, accessed on 2026-04-22, available at [https://www.figma.com/make/wDEXX00B4cynN04EYZDNWl/User-Journey-Map-Design?t=Fp0EfhNpg6uR8dC-1] Used for generating the format of "Figure 2 User Journey Map".
(2) Gemini 3 Flash, Google DeepMind, knowledge cutoff: dynamic (real-time training data up to 2026), accessed on 2026-04-22, available at [https://gemini.google.com/app/63e54a67c477750f?hl=zh-HK] Used for creative asset generation and UI design conceptualization.

[1] Biczók, G., Martínez, S. D., Jelle, T., & Krogstie, J. (2014). Navigating MazeMap: Indoor human mobility, spatio-logical ties and future potential. arXiv preprint arXiv:1401.5297.
[2] Dong, W., Wu, Y., Qin, T., Bian, X., Zhao, Y., He, Y., Xu, Y., & Yu, C. (2021). What is the difference between augmented reality and 2D navigation electronic maps in pedestrian wayfinding? Cartography and Geographic Information Science, 48(3), 225-240.
[3] Qiu, Z., Mostafavi, A., & Kalantari, S. (2023). Use of augmented reality in human wayfinding: A systematic review. arXiv preprint arXiv:2311.11923.

Reference